

ROBOTICS AND ARTIFICIAL INTELLIGENCE

Maximum Marks: 100

Time allowed: Two hours

1. *Answers to this Paper must be written on the paper provided separately.*
 2. *You will **not** be allowed to write during the first 15 minutes.*
 3. *This time is to be spent in reading the question paper.*
 4. *The time given at the head of this Paper is the time allowed for writing the answers.*
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5. *Attempt **all** questions from **Section A** and **any four** questions from **Section B**.*
 6. *The intended marks for questions or parts of questions are given in brackets[].*

Instruction for the Supervising Examiner

Kindly read aloud the Instructions given above to all the candidates present in the Examination Hall.

This paper consists of 12 printed pages.

SECTION A (40 Marks)

(Attempt *all* questions from this Section.)

Question 1

Choose the correct answers to the questions from the given options.

[20]

(Do not copy the questions. Write the correct answers only.)

- (i) Study the image given below. In which field is this robot used?



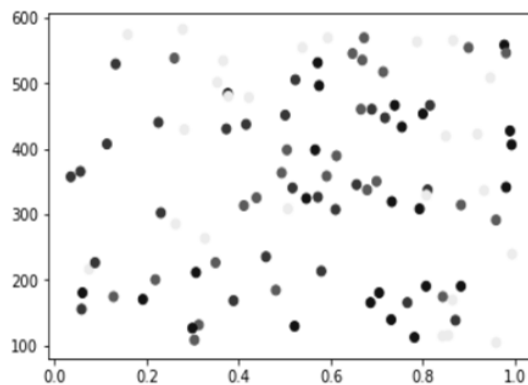
- (a) Healthcare
 - (b) Defence
 - (c) Warehouse management
 - (d) Entertainment
- (ii) Which of the following is an application of AI?
- (a) Face recognition
 - (b) Microwave ovens
 - (c) Room heater
 - (d) Geyser

- (iii) A Cobot is best described as a robot that:
- (a) mimics human behaviour
 - (b) collaborates with humans
 - (c) is used underwater
 - (d) is without sensors
- (iv) A revolute joint shows rotation along _____.
- (a) Multiple axis
 - (b) Double axis
 - (c) Triple axis
 - (d) Single axis
- (v) The gear that converts rotational motion to linear motion is:
- (a) Rack and pinion gear
 - (b) Spur gear
 - (c) Bevel gear
 - (d) Helical gear
- (vi) The actuator best suited for the rotational movement of a wheel is:
- (a) Hydraulic actuator
 - (b) Pneumatic actuator
 - (c) Electric rotary actuator
 - (d) Linear piston actuator

- (vii) _____ is a string function in Python.
- (a) upper()
 - (b) sort()
 - (c) append()
 - (d) extend()
- (viii) When a small spur gear turns a big spur gear, what happens?
- (a) The big gear turns faster and in the same direction.
 - (b) The big gear spins slower and in the same direction.
 - (c) The big gear spins slower and in the opposite direction.
 - (d) The big gear spins faster and in the opposite direction.
- (ix) The Python library for plotting graphs is:
- (a) NumPy
 - (b) Pandas
 - (c) Matplotlib
 - (d) SciPy
- (x) The main function of a control system is to:
- (a) provide power to enable the mobility of the robot only.
 - (b) process sensor data, make decisions, and act on it.
 - (c) design the robot's skeletal framework.
 - (d) store the information only.

- (xi) The main purpose of data modelling in Artificial Intelligence (AI) is to:
- (a) design user interfaces for better view.
 - (b) create algorithms that make predictions and discover patterns from data.
 - (c) store large amounts of data acquired from multiple reliable sources like Kaggle.
 - (d) visualise the data for better data evaluation.
- (xii) The main function of gears in a robotic system is to:
- (a) measure temperature.
 - (b) process sensor data.
 - (c) store code of programming language.
 - (d) transmit and amplify force.
- (xiii) Identify the feature of supervised learning.
- (a) Works with labelled data
 - (b) Works without labelled data
 - (c) Uses clustering techniques
 - (d) Works without any dataset
- (xiv) _____ is an ethical issue in computing.
- (a) Gear design
 - (b) Bevel gears
 - (c) Data theft
 - (d) Sensor calibration

- (xv) **Assertion (A):** Data acquisition is a stage of AI project cycle.
Reason (R): Without collecting data, a model cannot be trained.
- (a) (A) is true but (R) is false.
 - (b) (A) is false but (R) is true.
 - (c) Both (A) and (R) are true and (R) is the correct explanation for (A).
 - (d) Both (A) and (R) are true but (R) is not the correct explanation for (A).
- (xvi) The important factor to consider when collecting data for an AI project is:
- (a) data should be obtained from random websites on the internet.
 - (b) data should be authentic, reliable and accurate.
 - (c) data collection methods can be unethical and unlawful.
 - (d) private data can be accessed without permission.
- (xvii) Identify the type of plot shown below.



- (a) Line plot
- (b) Bar plot
- (c) Pie plot
- (d) Scatter plot

- (xviii) What is the primary benefit of using Tinkercad in robotics design?
- (a) To simulate and test robotic systems virtually.
 - (b) To program robots using Arduino.
 - (c) To build physical prototypes.
 - (d) To control robot movements.
- (xix) Function to count the number of items in a string 'a' in Python is _____.
- (a) `string.cnt ('a')`
 - (b) `count.string ('a')`
 - (c) `cnt.string ('a')`
 - (d) `string.count ('a')`
- (xx) Give the output of the following Python code:
- ```
print("A" * 2)
```
- (a) A2
  - (b) AA
  - (c) A22
  - (d) Error

## Question 2

Answer the following questions:

(i) Mention *any two* sensors used in autonomous cars. [2]

(ii) *"Autonomous delivery robots are being tested in cities to transport food and parcels directly to customers' doorsteps. These robots are equipped with sensors and cameras for navigation, and they help reduce delivery time and human effort."* [2]

Based on the above paragraph, mention *any two* benefits of using delivery robots in urban areas.

(iii) What is gear ratio? [2]

(iv) Describe how a machine will classify a raw mango from images of other fruits. [2]

(v) Mention *any two* methods of how a computer is used for committing cyber fraud. [2]

(vi) What is *supervised learning approach* in training AI models? [2]

(vii) Explain 'Who' and 'Why' in problem scoping. [2]

(viii) What will be the output of the following Python code? [2]

```
t = (1, 2, 3)
t = t + (4, 5)
print(t)
```

(ix) What will be the output of the following Python code? [2]

```
s = "Python"
print(s[:-1])
```

(x) What will be the output of the following Python code? [2]

```
sal = (7000, 4000, 11000, 1700)
print(max(sal))
print(min(sal))
```



## SECTION B (60 Marks)

(Answer **any four** questions from this Section.)

*The answers in this section should consist of the programs in either Python environment or any program environment with Python as the base.*

*Each program should be written using variable description / mnemonic codes so that the logic of the program is clearly depicted.*

*Flowcharts and algorithms are not required.*

### Question 3

- (i) State *any three* advantages of *Smart schools*. [3]
- (ii) Briefly explain Autonomous systems. Give *any two* examples. [3]
- (iii) Write a Python program to plot a line chart using *Matplotlib library* using the following data: [9]
  - 1. Import the necessary libraries.
  - 2. Store the dates from 1<sup>st</sup> September to 5<sup>th</sup> September (both inclusive) and their corresponding rainfall data as 90 mm, 92 mm, 180 mm, 120 mm, and 80 mm respectively.
  - 3. Plot a line chart to visualise the rainfall.
  - 4. Add appropriate labels for the X-axis (Date) and Y-axis (Rainfall).
  - 5. Set a title for the chart as 'Rainfall Data'.

#### Question 4

- (i) State *three* advantages of using cobots in industries. [3]
- (ii) Briefly explain Data Modelling. [3]
- (iii) Write a Python program to plot a *pie chart* comparing pollution ranking of five countries as given below: [9]

Country = ['Russia', 'India', 'China', 'Bangladesh', 'Nepal']

Pollution = [4, 2, 3, 1, 5]

Import the necessary *libraries*.

#### Question 5

- (i) Explain the difference between *contact-type sensors* and *non-contact type sensors*. Give an example for each. [3]
- (ii) Mention *any three* security measures to be taken to safeguard your computer from cyber threats. [3]
- (iii) Write a Python program that performs the following tasks: [9]
  - 1. Create a tuple having elements (100, 200, 700, 300, 150)
  - 2. Convert the above tuple to list.
  - 3. Remove the element 700 and in its place insert an element 650.
  - 4. Sort all the elements in ascending order.
  - 5. Convert the list to a tuple.
  - 6. Print the final tuple.

### Question 6

- (i) Describe briefly how a robotic arm with a revolute joint is controlled by a motor. [3]
- (ii) All robots are machines, but not all machines are robots. How can machines and robots be identified and distinguished from each other? [3]
- (iii) Write a Python program that performs the following operations on a string: [9]
1. Create a string with the value "Computer Applications"
  2. Find and print the length of the string.
  3. Count and print the occurrences of 'p' in the substring "Applications"
  4. Find and print the position of the substring "Applications" within the string.
  5. Check whether the string is in uppercase and print the result.
  6. Replace the substring "Applications" with "Science" and print the new string.

### Question 7

- (i) What role do vision sensors play in robotics? Give *one* example. [3]
- (ii) In the AI project cycle, what is the purpose of the data acquisition stage? [3]
- (iii) Write a Python program that performs the following task on a tuple: [9]
1. Create a tuple 't1' with elements (100, 110, 120, 330, 440)
  2. Concatenate a tuple 't2' with elements (77, 88) to 't1'
  3. Convert the above tuple to list.
  4. Append another list 'lst2' with elements [16, 26]
  5. Remove element in the second index.
  6. Insert element 40 to the list at index 5.
  7. Print the list.

### Question 8

- (i) List *any three* key components required to design a simple wheeled robot. [3]
- (ii) (a) What are the different ways of representing data? [3]
- (b) Name *any one* graph ideal for showing trends and changes in data over time.
- (iii) Write a Python program that does the following using string built-in functions: [9]
- stg = " Welcome to My Python Project "
1. Convert entire string to lowercase.
  2. Replace "my" with "your".
  3. Find the length of the new string.
  4. Check if the string ends with 'T'.
  5. Print the characters from index 3 to 6 in reverse order.